

BEYOND THE GATE

Translating Research into Societal Impact and Commercialization

Keynote Address

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5 August 2026

For so long, universities have been regarded as ivory towers — towers that stand tall above all others in knowledge, in morals, in values, in ethics, and in discipline.

The dictionary defines an ivory tower as *a state of privileged seclusion or separation from the facts and practicalities of the real world*. The term dates back to the nineteenth century, coined to describe an environment of intellectual pursuit disconnected from the practical concerns of everyday life — and today, it is used almost exclusively to describe academia itself: the college and university system.

I find that definition uncomfortable. Not because it is unfair, but because it is precise. Privileged. Secluded. Separated from the facts and practicalities of the real world. Is that what we have built? Is that what we intend to remain?

But a tower is only as strong above the ground as it is deep below the ground.

So I ask: how deep are our roots? How deeply are we connected to the ground we stand on? Are we rooted in the challenges that make up our societies, or are we rooted in challenges imported from abroad? Are we in touch with the realities that make up our localities? Are we aware of the problems we were designed and built to solve?

If the answer is no — or not very certain — then we must revisit our foundation. What must we be doing wrong?

Distinguished Vice Chancellor, Deputy Vice Chancellors — particularly our host, the DVC Research, Innovation and Strategic Partnerships — Provost, Deans, Professors, Captains of Industry, particularly the Director General of the Development Agenda for Western Nigeria and his amazing team, members of the organizing committee, distinguished ladies and gentlemen — permit me to stand on existing protocols.

I am humbled to be saddled with the responsibility of this keynote address. This university has played a significant role in my life, as I am sure it has in all of ours. It has done — and still does — so much that it is globally recognized and certified as the best university in Nigeria. That position alone is enough to make us feel the weight of Nigeria's developmental challenges, because it means: if Nigeria must get it right, then we here at the University of Ibadan must lead all others in doing it right.

So the question of how deep our roots should be is exactly the gap we are gathered here to address.

The gap we are discussing today is not an abstract one. It is the gap between a hungry population and the food security research sitting in our labs. It is the gap between a nation anxious about insecurity and the surveillance, robotics, and monitoring solutions our engineering departments could deploy tomorrow if given the chance. It is the gap between millions of young, unemployed graduates and the industries that could be built around research this university produces every single year. Our population is growing faster than our job creation. Our insecurity is growing faster than our deployed solutions. And so "Beyond the Gate" is not a metaphor for today — it is an emergency.

Beyond the Gate

The theme has been carefully crafted, and its words are worth unpacking.

"**Beyond the Gate**" calls us to take research to the people who need it — to the solutions of those in remote areas. And when we say "take it to them," we are not talking about what happens *after* the research is done. No. Even before the research begins — at the point of identifying the problem — how connected are we to those outside the gate? At the point of methodology, when we are choosing our materials, our approach, our assumptions about

reality — how well are we listening to the environment we claim to serve? And at the point of testing and feedback, how much do we rely on the very people the solution is meant for?

Beyond the gate means the people outside are just as relevant as any stakeholder inside the academy — because it is their insight that should shape both the problems we solve and the way our solutions are adopted.

"Translating research into societal impact and commercialization" forces us to confront the question of *sustainability*. Are we building solutions with enough impact on people's lives that communities themselves would fight to keep them alive — solutions people would fund, protect, and sustain? Would they be commercially viable?

These two things — social impact and commercial viability — are not always the same thing. Sometimes they run in parallel. Sometimes, sadly, they are mutually exclusive.

Take **security**. Everyone in this country is worried about insecurity. But tell people to *pay* for the security measures that protect them, and they will ask you: why should we pay? Isn't that what government is for? Commercial viability may not be there — but social impact certainly is.

Now flip it. Think of something with strong commercial viability but little or no positive social impact — take **sports betting and gambling platforms**. They are built by some of our sharpest local tech talent. They generate enormous revenue. They are, without question, commercially successful. But do they build the society we want? Or do they quietly extract value from the very communities they operate in, offering little back in return?

This is the tension "Beyond the Gate" asks us to resolve — solutions that are both wanted *and* worth wanting. It is do-able. Cast our minds to history and see myriads of examples.

Two Stories

Let me ground this in two stories.

The first is the solar panel. It did not begin as a commercial product. It began as research — decades of quiet, unglamorous work on the photovoltaic effect, refined slowly in labs with no guarantee of market success. It took deliberate translation — investment, manufacturing partnerships, policy support, falling costs — to move it from a laboratory curiosity into one of the most impactful commercial technologies of our time. And nowhere is that more visible than here in Africa, where solar has leapfrogged grid electricity for millions of people, and where it now quietly powers something just as vital: the preservation of food. Solar-powered cold storage for farmers — keeping tomatoes, vegetables, and perishables from rotting before they ever reach a market — is one of the most powerful examples of research crossing the gate into societal impact.

The second story is more personal. Some time ago, I attended GITEX Global in Dubai — one of the world's largest technology exhibitions. There, I met a young Chinese marketer who showed me a device that, quite frankly, swept me off my feet. He explained that it monitored the temperature of vegetables — and by extension, any perishable goods — as they moved through courier delivery. If the temperature fell below, or rose above, the required safe range at any point in the journey, the device would log it and update a web platform in real time, alerting whoever owned the shipment.

Think about why this matters. Food, medicine, even organs for transplant can pass through unsafe conditions during delivery — and by the time they arrive, they can *look* perfectly fine. You would have no way of knowing they had been compromised. You would assume they had been kept in good condition throughout the journey, when in fact they had not.

Technically, this is not a complicated solution. A simple Wi-Fi-enabled microcontroller, a temperature sensor, GPS if you like, streaming data to the cloud — any developer could build a web or mobile app around that. This was not a breakthrough in research. It was a breakthrough in **translation**. A young marketer sold it to me as a ready, working, revenue-generating product already in active use.

That is the difference between research done for its own sake, and research carried all the way through to commercialization for social impact. We so often do the hard part — the invention — and stop halfway, because we are already chasing the next idea, never pausing to finish and sell the one in hand.

Seven Ingredients for Crossing the Gate

If we are to consistently take research beyond the gate, I believe there are seven ingredients that must be in the mix.

1. Creative Ideas.

Everything begins here. It is the audacity of an idea — the willingness to imagine something that does not yet exist — that gives us anything worth building or marketing at all. Innovation before infrastructure.

2. Tech Knowledge and Talent.

Once the idea exists, it needs hands and minds capable of building it — deep technical knowledge, engineering capacity, and the discipline to turn concept into working product.

3. Trust in Local Talent.

It is not enough for the talent to exist — we must *trust* it. Too often, industry and government reach for imported solutions and expatriate expertise before they have even tested what is available locally. Beyond the gate begins with believing in what is built inside it.

4. Marketing.

Nobody is going to come begging for your solution. Remember the young man in Dubai — the technology he sold me was not extraordinary; the *selling* was. We are, frankly, weak at this. We build and disappear. We must learn to package, present, and sell our research with the same energy we put into researching it.

5. Regulation.

Regulation is not the enemy of innovation — it is often its engine. If nobody forces you to carry an umbrella, you probably won't. In the UK, fire safety regulations are taken seriously — not because people are naturally cautious, but because insurance companies and the law compel it. That pressure creates a *market* for solutions. Good regulation, properly designed, is what makes society demand innovation rather than resist it.

6. Partner with Labs — the Triple Helix.

Researchers are, by nature, always looking for the next gap, the next improvement, the next version. Left alone, a research idea may never feel "ready" for market. This is why we need deliberate partnerships — a triple helix of government, research institutions, and private industry — where a company can take existing research, build a minimum viable product, and carry it the rest of the way to deployment.

7. Design for Self-Sustaining Social Impact.

Too many research ideas are built with one eye on the next grant — from an NGO, a government agency, a foundation. But donor interest shifts. What must be built instead are solutions with enough societal value that the society itself will not let them die — solutions that pay for themselves through the value they create, not through the goodwill of a funder. This, again, comes back to creativity: designing solutions people will not want to live without.

The Fuel Robotics and AI Cannot Run Without: Data

Speaking from the perspective of my domain expertise, I can say to you that every robotics solution, every AI model, every "intelligent" system we are proud to demonstrate at conferences like this one runs on one fuel: **data**. And I have sat in enough rooms with researchers to hear the same lament again and again — there is no data. No local, authentic, real-time data to train on, to validate against, to build with. So what happens? We reach for foreign datasets — data harvested from other people's roads, other people's markets, other people's bodies, other people's weather — and we build "local" solutions on foreign foundations. Ask yourself what that produces. A robotics system trained on someone else's reality will always be a stranger in ours.

This is not merely a technical inconvenience — it is a national vulnerability. If our researchers cannot access authentic Nigerian data, our AI cannot be relevant to Nigerian problems. Government must wake up to this. Private industry must wake up to this. And the encouraging news is that we are not starting from zero — Nigeria already has the **NDPR, the Nigeria Data Protection Regulation**, a policy framework for exactly this kind of governance. What we now need is the deliberate work of building **data exchange platforms** — safe, regulated, authentic pipelines between industry, government, and researchers — so that

the data our institutions hold is not locked away, but responsibly shared to fuel the very solutions meant to serve us.

Robotics and AI will not save us on imported intelligence. They will save us on Nigerian data, gathered, governed, and shared with integrity.

Building the Bridge: What University-Industry Partnership Must Look Like

The letter inviting me here asked that I speak to the imperative of sustainable university-industry partnerships, and to the strategic approaches required to build them. Allow me to be specific.

Industry must invest in laboratories. Not as a donation for public relations, but as infrastructure for their own future pipeline of solutions. A well-equipped lab, donated or co-funded by industry, gives researchers the ability to simulate, test, and validate against real-world, real-time data — the very thing that is currently missing.

Industry and government must sponsor field-specific conferences and forums — the kind of focused, town-and-gown conversations I mentioned earlier — so that ideas do not stay trapped in departments but meet the people who can fund and deploy them.

And industry must invest in our PhDs. This is where the triple helix truly earns its name. A PhD researcher is, by definition, someone spending years going deep into a single unresolved problem. If that problem is sponsored by an industry partner — shaped around a real, live challenge — the university produces not just a thesis, but a solution with a customer already waiting.

I saw this work firsthand in the United Kingdom, through a structure called the **Knowledge Transfer Partnership — KTP**. Under this model, a company brings a genuine, unsolved problem to a university. A researcher — often a PhD-level associate — is embedded inside that company, funded jointly by government, the company, and the university, to work on the problem in real time. The company gets a resident expert and a working solution to a live business challenge. The university gets a researcher testing theory against reality. And the researcher gets to build something that matters, backed by both institutions. Everybody wins. This is not a foreign luxury we cannot replicate — it is a template we can localize, today, between our companies and this university.

This is what sustainable university-industry partnership looks like: not a one-off donation, not a photo opportunity, but laboratories, forums, and PhDs — deliberately, structurally, tied to real industry problems.

The Challenges Before Us

None of this is simple. Three challenges stand squarely in our way.

IP Protection. Too often, research findings leak out before they are protected — quietly reproduced, tweaked, and repackaged elsewhere, sometimes better than we could have done ourselves, simply because we did not secure what was ours in time.

Brain Drain. This is not only about people leaving the country — it is often about our best minds staying *physically* in Nigeria while working, in effect, for competitors abroad. Foreign companies quietly benefit from work that should have been deployed here. It gives us a false sense of contribution while robbing us of the time and talent we need for national development.

Funding. Research tools are expensive. Researchers abroad often work with heavily subsidized software, hardware, and platforms — subsidies we do not have here. Without them, our researchers cannot fully test, simulate, or validate their work before it reaches industry — and industry, in turn, rejects our solutions as "not tested in real conditions."

This is precisely where **Artificial Intelligence** can help close the gap. AI now makes it possible to build simulated environments that replicate real-world deployment conditions — allowing our researchers to test, refine, and validate solutions locally, at a fraction of the traditional cost, before they ever leave the lab.

One structural response to all three of these challenges is **local, field-specific town-and-gown forums** — not broad, general conferences, but focused sessions where, say, researchers in accounting sit with accounting firms ready to implement. Where agricultural researchers sit with agribusinesses. Field by field, so the conversation stays sharp and the outcomes stay actionable.

I would also point us to existing legal instruments — laws like the **National Office for Technology Acquisition and Promotion (NOTAP)** — designed precisely to protect our intellectual property and our professionals. These laws exist. What we lack is the culture of using them consistently.

In Conclusion

As a nation, we are gifted. We are wealthy, and our potential shows itself daily — in our universities, in our societies, in international competitions and international arenas. The quality of our tutors, lecturers, and professors, wherever they go abroad, is consistently outstanding. There is no doubt that the capacity is within us.

What we do not have enough of are the **structures** that transport research out of the university and into society — structures for deployment, structures that turn knowledge into production.

It is particularly painful when organizations claim our research does not fit the realities of deployment, and so bring in expatriates instead — expatriates who, more often than not, decline to transfer their knowledge. The result is a cycle: we invite them again and again for the same problems, never building the local capacity that would let us solve it ourselves.

Today's forum gives us the opportunity to confront this honestly, and to commit ourselves to something better. I am confident that the quality of the conversations we will have here can help improve the quality — and the deployment — of local research.

We must unlearn, learn, and relearn one central truth: **a society can only do better when it deploys solutions researched and built to fit its own environment.** In this, the university must always be a critical player.

Let us rally round — to partner, to support, and to carry this mission beyond the gate.

Thank you for listening, and God bless you all.